

FA 9 in Statistical Theory

1.

Given:

Observed: 24, 108, 95, 23

Expected: 30, 90, 90, 30

Test Statistic:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Computing each term:

$$\frac{(24 - 30)^2}{30} = 1.20$$

$$\frac{(108 - 90)^2}{90} = 3.60$$

$$\frac{(95 - 90)^2}{90} \approx 0.28$$

$$\frac{(23 - 30)^2}{30} \approx 1.63$$

$$\chi^2 \approx 6.71$$

Degrees of Freedom:

$$df = 4 - 1 = 3$$

Critical Value: $\alpha = 0.05$, $df = 3$

$$\chi_{crit}^2 = 7.815$$

Since $6.71 < 7.815$, we **fail to reject** H_0 . There is no significant evidence that the coin is unfair.

2.

Observed Table:

	Passed	Failed	Total
Class A	72	17	89
Class B	68	23	91
Total	140	40	180

Expected Value Formula:

$$E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

Expected Values:

$$E_{A,\text{passed}} \approx 69.22$$

$$E_{A,\text{failed}} \approx 19.78$$

$$E_{B,\text{passed}} \approx 70.78$$

$$E_{B,\text{failed}} \approx 20.22$$

$$\chi^2 \approx 1.05, \quad df = 1$$

Critical values:

$$\alpha = 0.05 \rightarrow 3.841, \quad \alpha = 0.01 \rightarrow 6.635$$

Since $1.05 < 3.841$, we fail to reject H_0 .

(b) With Yates correction

$$\chi^2 \approx 0.96$$

$0.96 < 3.841$, so we still fail to reject H_0 .

Therefore, there is no significant difference between Class A and Class B.

3.

Expected Values:

$$E_{11} = \frac{74 \cdot 79}{200} \approx 29.23$$

$$E_{12} \approx 44.77, \quad E_{21} \approx 49.77, \quad E_{22} \approx 76.23$$

Observed Table:

	Blonde	Not Blonde	Total
Blue	49	25	74
Not Blue	30	96	126
Total	79	121	200

Chi-square components:

$$\begin{aligned}\frac{(49 - 29.23)^2}{29.23} &\approx 13.37 \\ \frac{(25 - 44.77)^2}{44.77} &\approx 8.73 \\ \frac{(30 - 49.77)^2}{49.77} &\approx 7.85 \\ \frac{(96 - 76.23)^2}{76.23} &\approx 5.13\end{aligned}$$

$$\chi^2 \approx 35.08$$

(a) Coefficient of Contingency

$$C = \sqrt{\frac{\chi^2}{\chi^2 + N}} = \sqrt{\frac{35.08}{35.08 + 200}} \approx 0.39$$

With Yates correction:

$$\chi^2 \approx 33.33$$

(b) Maximum Coefficient of Contingency

$$C_{\max} = \sqrt{\frac{k-1}{k}}, \quad k = 2$$

$$C_{\max} = \sqrt{\frac{1}{2}} \approx 0.707$$

Since 0.39 is moderately close to 0.707, there is a moderate association between hair color and eye color.

4.

(a) Model: $mpg = B_0 + B_1 wt$

Fitted line:

$$\hat{mpg} = 37.885 - 5.344 wt$$

(b) Model: $wt = a_0 + a_1 mpg$

Fitted line:

$$\hat{wt} = 6.047 - 0.141 mpg$$

(d) Interpretation:

For the model of mpg on weight, for every 1 unit increase in weight (1000 lbs), mpg decreases by about 5.34. This means heavier cars are less fuel efficient.

For the model of weight on mpg, for every 1 unit increase in mpg, weight decreases by about 0.141 (1000 lbs), meaning more fuel-efficient cars tend to be lighter.